

Low-Level Design (LLD)

RAG-Based Customer Support Assistant

1. Module-Level Design

Module	File	Responsibility
Main Interface	main.py	Parses CLI commands, supports ingest, ask, demo, and interactive mode.
Pipeline Coordinator	src/rag_pipeline.py	Connects loader, chunker, embeddings, retriever, router, HITL, and workflow.
Document Loader	src/loader.py	Loads PDF files and extracts page text.
Chunking Module	src/chunker.py	Splits extracted pages into overlapping chunks with metadata.
Embedding Module	src/embeddings.py	Creates deterministic local embeddings.
Retriever	src/retriever.py	Indexes chunks in ChromaDB and retrieves top-k matches.
Router	src/router.py	Classifies questions into customer support intents.
HITL Module	src/hitl.py	Applies escalation rules.
Workflow	src/workflow.py	Defines LangGraph nodes, edges, and state transitions.
Answer Generator	src/llm.py	Generates final answers from retrieved support content.

2. Data Structures

DocumentPage

- source_id: unique page identifier
- file_name: PDF file name
- file_path: path to source PDF
- page_number: source page number
- text: extracted page text

Chunk

- chunk_id: unique chunk identifier
- source_id: parent page identifier
- text: chunk content
- metadata: file name, file path, page number, and chunk index

RetrievalHit

- chunk_id: matching chunk ID
- text: retrieved text
- metadata: source metadata
- distance: vector search distance

RouteDecision

- route: detected intent category
- confidence: routing confidence
- reason: explanation for routing decision

3. Workflow Design (LangGraph)

Node	Purpose
process	Routes the query, retrieves matching chunks, and checks HITL rules.
human_review	Creates an escalation response when human review is required.
output	Returns the final answer and source metadata.

Nodes:

- Input Node → receives query
- Router Node → detects intent
- Retriever Node → fetches chunks
- HITL Node → checks escalation conditions
- Generator Node → generates response
- Output Node → returns result

Edges (Flow):

Input → Router → Retriever → HITL → Generator → Output

State Transitions:

- Query enters system → intent assigned
- Context retrieved → confidence evaluated
- Decision taken → response generated OR escalated

Graph Edges

- START -> process
- process -> output when automated response is safe
- process -> human_review when escalation is required
- human_review -> output
- output -> END

4. Conditional Routing Logic

Route	Example Keywords	Purpose
orders_delivery	order, track, delivery, shipping, address	Handles order tracking and delivery questions.
billing_payments	payment, charged, invoice, refund, return	Handles billing, invoice, refund, and return questions.
account_management	account, password, login, sign up, email	Handles account and login support.
technical_support	app, website, slow, loading, browser, cache	Handles technical troubleshooting.
escalate	fraud, legal, lawsuit, chargeback, complaint, vip	Routes sensitive cases to human review.
needs_clarification	very short or vague input	Requests human or clarification path.

Answer Generation Criteria:

- High confidence score
- Relevant chunks retrieved

Escalation Criteria:

- Low confidence
- Sensitive keywords (legal, complaint, fraud)
- No relevant chunks

5. HITL Design

HITL escalation is triggered when automation may be unsafe or unreliable. The system escalates when the route is sensitive, the question is too vague, no chunks are found, retrieval confidence is low, or the query is too complex.

When Escalation is Triggered:

- Low retrieval confidence
- Ambiguous queries

- Sensitive requests

What Happens After Escalation:

- Query is flagged
- Response indicates human support required
- (Optional) Sent to ticketing system

Benefits:

- Improves reliability
- Prevents incorrect answers

Limitations:

- Increases response time
- Requires human availability

6. Interface Design

The assistant supports three main run modes:

- `python main.py ingest`: build the ChromaDB index
- `python main.py`: start interactive question-answer mode
- `python main.py ask "question"`: ask a single question directly

7. Error Handling

- Missing data directory raises `FileNotFoundError`.
- No PDF files raise `ValueError`.
- No matching chunks triggers escalation.
- Low confidence triggers human review.
- The offline answer generator keeps the system runnable without an API key.